Quiz 2

Name: _____ Student ID: _____ Marks:_____

1. Evaluate $f(z) = \oint_\gamma \sin\left(\frac{1}{z^2+z-2}\right)\,\mathrm{d}z$, where γ is a circle $|z-1+2\mathrm{i}| = \frac{3}{4}$. [7]

2. Evaluate the integral $\oint_\gamma \dfrac{1}{z-1}\,\mathrm{d}z$, where γ is the circle of radius 2, centred at the origin. [5]

3. Evaluate the integral $\oint_\gamma \dfrac{3z^2 \cosh z}{(z+2\mathrm{i})^2}\,\mathrm{d}z$, where γ is the circle of radius 8, centred at 1. Find the real and imaginary parts of the integral you have evaluated. [8]